

Score function space transformation Technique

Derivation Based on Differentiability and Chain Rule

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1 Variable Definition

$\mathbf{X}$: original data vector

r_i : scaling factor($i = 1, 2, \dots, 10$)

d_1 : mean value of original data

d_2 : design variable by scaling original data

2 Change of variable Technique

$$\mathbf{d}_2 = \text{diag}[r_1, r_2, \dots, r_N] \mathbf{d}_1$$

$$g_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_2) = f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_1) \cdot |\mathbf{J}|$$

$$* |\mathbf{J}| = \left| \frac{\partial \mathbf{d}_1}{\partial \mathbf{d}_2} \right| = \begin{vmatrix} \frac{\partial \mathbf{d}_{1,1}}{\partial \mathbf{d}_{2,1}} & \frac{\partial \mathbf{d}_{1,1}}{\partial \mathbf{d}_{2,2}} & \dots & \frac{\partial \mathbf{d}_{1,1}}{\partial \mathbf{d}_{2,10}} \\ \frac{\partial \mathbf{d}_{1,2}}{\partial \mathbf{d}_{2,1}} & \frac{\partial \mathbf{d}_{1,2}}{\partial \mathbf{d}_{2,2}} & & \\ \vdots & & \ddots & \\ \frac{\partial \mathbf{d}_{1,10}}{\partial \mathbf{d}_{2,1}} & & & \frac{\partial \mathbf{d}_{1,10}}{\partial \mathbf{d}_{2,10}} \end{vmatrix} = \begin{vmatrix} \frac{1}{r_1} & & & \\ & \frac{1}{r_2} & & \\ & & \ddots & \\ & & & \frac{1}{r_{10}} \end{vmatrix} = \left| \frac{1}{r_1 \cdot r_2 \cdots r_{10}} \right|$$

$$g_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_2) = \frac{1}{r_1 \cdot r_2 \cdots r_{10}} \cdot f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_1)$$

$$\log g_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_2) = - \sum_{i=1}^{10} \log r_i + \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_1)$$

$$\frac{\partial}{\partial d_{2i}} \log g_{\mathbf{X}}(\mathbf{X}; \mathbf{d}_2) = \frac{\partial}{\partial d_{2i}} \left(\sum_{i=1}^{10} -\log r_i \right) + \frac{\partial}{\partial d_{2i}} \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}_1)$$

$$* \mathbf{r} = \mathbf{d}_2 \oslash \mathbf{d}_1, \quad r_i = \frac{d_{2,i}}{d_{1,i}}$$

1st. term

$$\frac{\partial}{\partial d_{2,i}} (-\log r_i) = \frac{\partial}{\partial d_{2,i}} \left(-\log \frac{d_{1,i}}{d_{2,i}} \right) = \frac{\partial}{\partial d_{2,i}} (-\log d_{1,i} + \log d_{2,i}) = \frac{1}{d_{2,i}}$$

3 Example

- Binomial gaussian distribution

$$f(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

$$\boldsymbol{\mu} : \text{평균 벡터} [\mu_1, \mu_2]^\top$$

$$* \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

- score function w.r.t. $\mathbf{x}$

$$\log f(\mathbf{x}) = \log\left(\frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}}\right) + \log\left(\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)\right)$$

$$\log f(\mathbf{x}) = \underbrace{-\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Sigma}|}_{\text{Constant Term (w.r.t. } \mathbf{x})} \underbrace{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}_{\text{Quadratic Term}}$$

$$\nabla_{\mathbf{x}} \log f(\mathbf{x}) = -\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) = \left[-\frac{x_1 - \mu_1}{\sigma_1^2}, -\frac{x_2 - \mu_2}{\sigma_2^2} \right]^\top$$

- score function w.r.t. $\boldsymbol{\mu}$

$$\ln f(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \ln\left(\frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}}\right) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

- Binomial lognormal distribution